

Distributed CubeSat mission concepts

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Alessandro Golkar

Skolkovo Institute of Science and Technology, Center for Entrepreneurship and Innovation, Moscow, Russia

1 Introduction

Instead of relying on individual spacecraft, distributed CubeSat systems (DCS) allocate the required mission functionality across multiple flight units. A distribution strategy allows mission designers to enable new mission capabilities such as ubiquitous connectivity at costs compatible with commercial intents or enhance system features of interest such as temporal resolution and interspacecraft-coordinated operations. Distributed spacecraft design strategies are indeed relevant for spacecraft platforms of all sizes, though they are of special interest to CubeSat systems due to their inherent enabling features. Distribution allows CubeSats to achieve mission performance or mission assurance levels that would be more complex or even not possible to achieve by an individual nanosatellite system. It allows for new design approaches to be put in place, such as “fail fast, fail often” strategies that have been historically at odds with the design philosophy of spacecraft systems but that are now being actively experimented in private space ventures. Several spacecraft distribution strategies are possible, each of which offering different opportunities while exhibiting inherent technical challenges. This chapter offers an overview of distributed CubeSat mission concepts and architectures that are presently flying or have been proposed and that could be thus envisioned for future mission implementations. Satellite distribution offers several avenues for innovation, as several distribution strategies and related technologies have not yet been explored in the domain of CubeSats.

This chapter is structured as follows. [Section 2](#) provides a comprehensive analysis of feasible concepts of DCSs and a synthesis of the state-of-the-art distributed CubeSat mission concepts and architectures. [Section 3](#) discusses enabling technologies for DCS in the technical domains related to onboard autonomy and spacecraft coordination. [Section 4](#) discusses conclusions from the analysis of DCS and outlines a potential way forward for the adoption of those concepts in the emerging new space industry worldwide.

2 Distributed CubeSat system concepts

2.1 State of the art

The notion of DCS encompasses all CubeSat missions that involve more than one spacecraft unit to fulfill mission requirements. As such the notion of DCS is quite broad and includes an array of well-known concepts such as satellite constellations, formation flight missions, and satellite swarms. It also includes innovative concepts that have not flown yet at the time of writing of this chapter, such as fractionated and federated CubeSat systems. While distributed concepts have been implemented in several occasions in “standard” missions employing traditional “large” spacecraft, they are still unexploited in the domain of CubeSats. The most notable examples as of today, in terms of number of spacecraft flown, are the CubeSat constellations operated by Planet [1] and Spire [2], which account for 355 and 103 CubeSats, respectively, that are currently operating in orbit (according to public database sources, at the time of writing [3]).

There are several examples of DCSs that have already been operated in orbit. Referring to previous work by Selva et al. [4], it is possible to classify distributed satellite concepts from an architectural standpoint, referring to five fundamental factors that can either take high (H) or low (L) values for the given concepts. These are *homogeneity*, *size*, *spatial separation*, *functional interdependence*, and *operational independence*. The definition of those factors is provided in Table 1.

Selva’s taxonomy identifies six distributed satellite concepts using the proposed factor classification, mapping them to the known distributed concepts of satellite constellations, swarms, clusters, fractions, trains, and federations (Table 2). It also identifies potential other concepts through morphological analysis, which however do not map to any well-known concept formulation. We analyze each concept individually and discuss challenges and opportunities specific to CubeSat implementations.

Table 1 DCS classifying factors’ definition.

Homogeneity	The degree of structural similarity of the units
Size	Number of units participating into the DCS
Spatial separation	Maximum physical distance between the units in the DCS
Functional interdependence	The degree to which each unit in the DCS relies on other units to perform basic functions
Operational independence	The degree to which each unit is operated and/or owned by different organizations

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Table 2 Distributed CubeSat system concepts.

Name	Homogeneity	Size	Spatial separation	Functional interdependence	Operational independence
Cluster	H	L	L	H	L
Constellation	H	H	H	L	L
Federation	L	H	H	L	H
Fractionated	L	L	L	H	L
Series	H	L	H	L	H
Swarm	H	H	L	L	L
Train	L	L	L	L	H

2.1.1 Constellations

CubeSat constellations are currently being operated in orbit, such as those previously mentioned operated by Planet and Spire. Forty-one additional constellations are planned to be launched or proposed to be developed in the coming years [3]. Fig. 1 provides the most recent statistics of CubeSat constellations. Constellations are distributed systems made of identical flying units.

As for non-CubeSat counterparts, CubeSat constellations find use across a wide range of applications. Such applications benefit from an improved temporal resolutions and ubiquitous availability of service when compared with traditional

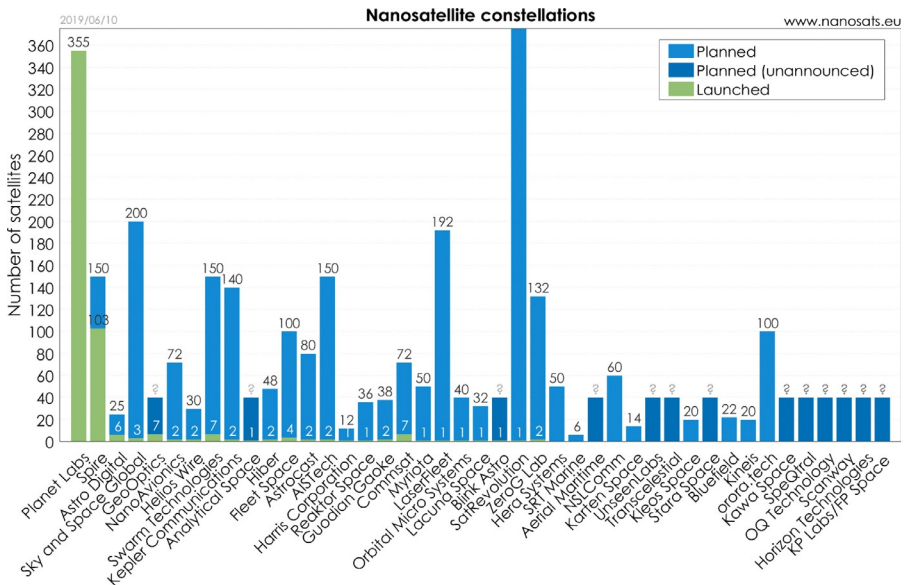


Fig. 1 Nanosatellite constellations.
From E. Kulu, Nanosats Database. www.nanosats.eu (Accessed 4 September 2019).

“monolithic” systems. Applications include Earth imaging, weather data, satellite automatic identification systems (AIS), space-based Automatic Dependent Surveillance-Broadcast (ADS-B), and Internet of Things (IoT) connectivity.

CubeSat constellations are finding increasing commercial relevance due to the advancement in miniaturization of technology and increasing power utilization efficiency of radiofrequency components. Missions that were unconceivable few years past have been now demonstrated on miniaturized platforms. For instance, Sky and Space Global has demonstrated the first voice call routed through a CubeSat-based distributed system [5].

As it can be expected, DCSs will hardly replace large satellite platforms in applications requiring high-end performance or capacity. However, they represent a viable trade-off in terms of value for money and can be seen as complementary assets enhancing services based on data derived from “traditional” larger spacecraft. Such complementary includes the opportunity for fusing low to mid fidelity data acquired at a higher temporal resolution (through DCS) with high-fidelity data from larger platforms for improved mission readiness or quality of service. An example application of such opportunity is represented by coordinated Earth observations, fusing optical and synthetic aperture radar (SAR) data to inform mission scheduling and operations. While institutional constellations of high-fidelity instrumentation cannot be scaled to large numbers for cost reasons, small instrumentation can be scaled and used to improve the utilization efficiency of larger assets. From a commercial perspective, CubeSat constellations provide the opportunity to deliver space-based services at a cost per unit service that is one or two orders of magnitude lower compared with legacy systems.

Challenges in the development of future CubeSat constellations are twofold: economic and technical. From an economic perspective, CubeSat operators are still to demonstrate the long-term financial viability of their value proposition. While 43 CubeSat constellation proposals have appeared over the last decade, only Planet and Spire have delivered major constellations that are currently operational in orbit. All other operators have launched minimum viable technology demonstrators made of one to seven flying units. Those demonstrators are typically designed to retire technical, financial, and other types of risks in the project and to secure funding in successive fundraising rounds. A consolidation of CubeSat constellation proposals is therefore likely to be expected in the coming 5-year timeframe. From a technical point of view, a major challenge to overcome is the saturation of the radiofrequency spectrum due to exponentially larger utilization. The emergence of CubeSat constellations increased the number of orbiting spacecraft to unprecedented volumes compared with the near past. Licensing for spectrum is a cumbersome process taking up to 3 years to complete. The challenge is further complicated when dealing with constellations operating intersatellite links, due to the licensing constraints related to spacecraft coordination, noninterference with geostationary assets, and shared spectrum utilization. Such limitations will drive CubeSat operators to push for the development of transceivers at higher frequency or rely on optical communication solutions for their cross-link and/or downlink communication needs. Many of those solutions are currently under development at the time of writing of this chapter [6–8]. They will in turn

entail challenges in terms of size, weight, and power (SWaP); pointing accuracy; and increased payload complexity.

Large numbers of CubeSats flying in orbit will also drive for the development of new solutions for space situational awareness, space debris, spacecraft autonomy, and on-orbit traffic management. Spacecraft coordination will in particular play an important role in the development of advanced CubeSat instrumentation, which impacts research and development in artificial intelligence (AI)-enabled decision-making, power-efficient flight avionics, advanced telecommunication protocols, and proximity flight operations, among other relevant technical areas.

2.1.2 *CubeSat clusters, series, swarms, and trains*

The literature on distributed satellite systems does not have yet a universally accepted nomenclature of all possible distributed mission concepts. Several classifications have been proposed to classify distributed satellite missions from different perspectives (such as from architectural, functional, operational, and other perspectives). The divergence in definitions is particularly wide in the discussion of swarms, series, clusters, and trains. In this chapter, we adopt the taxonomy of Selva et al. [4]. Their paper provides references to other classification systems. For these reasons, some of the concepts discussed in this chapter may be classified under different names rather than those defined by the original proponents.

CubeSat clusters are DCS made of a low number (typically 2–6) of identical flight units, flying at low along-track distance between each other while being mutually dependent from a functional and operational perspective to deliver on their mission goals. An example of a satellite cluster in the “traditional” domain is the TechSat 21 mission concept [9]. Satellites in a cluster can simulate the activity of larger monolithic satellites. In remote sensing, for example, a satellite cluster can be designed to reproduce large effective apertures by using sparse array interferometry and synthetic aperture techniques [10]. In the communication domain, torrent-like architectures have been proposed for increasing downlink and uplink capacity of CubeSat platforms, typically constrained by power, volume, and geometry [11]. In the CubeSat domain, no cluster mission has flown to date (to the best of this author’s knowledge). Several concepts have been proposed and are under development at the time of writing of this chapter. CubeSat-enabled cluster concepts are, for example, those proposed for bidirectional reflectance distribution function (BRDF) measurements [12]. Another cluster concept is that of the SunRISE mission currently under development at NASA JPL for improving the understanding of the acceleration of solar energetic particles (SEPs) in coronal mass ejection (CME) events using space-based interferometry [13].

CubeSat series are distributed concepts where identical (or quasi-identical) units are flown, usually for technology demonstration purposes such as demonstration of inter-satellite link or formation flight technology. For example, the AeroCube-4 series of three nanosatellites has among its objectives the demonstration of formation flight through differential drag strategies, using deployable solar arrays [14]. CubeSat series are quite common and flown, mostly as “*proof-of-concept*” technology demonstrators.

Closely tied to the concept of cluster is the DCS concept of swarm. *CubeSat swarms* differ from clusters in that they involve a large number of flight units that are functionally noninterdependent from each other. Swarms offer the advantage brought in by large numbers of spacecraft, wherein mission performance can be designed to degrade more gracefully in case of failure of an individual flight unit. CubeSat swarms have not flown to date; most notably, FemtoSats (100-g class) have been proposed in swarm implementations for synthetic aperture applications (1000 or more flight units) [10]. It is unlikely, but not impossible, that large swarms of CubeSats (1000+ units) will fly in the near future. The typical application of swarms is of scientific nature. As they entail large numbers, swarms bring significant technology challenges in terms of spacecraft in-orbit identification and discrimination after deployment, coordination, and relative navigation.

CubeSat trains are DCS of different spacecraft, flying in coordinated orbits while performing independent operations. Distributed concepts such as NASA's A-Train [15] use large monolithic spacecraft to take independent observations that are later fused in ground postprocessing activities. CubeSat trains have not yet appeared in orbit, but they are likely to emerge as a natural evolution of the small satellite industry. As CubeSat instrumentation is becoming of interest for operational purposes, CubeSat trains become ideal complements of large institutional programs such as the European Copernicus [16].

2.1.3 Fractionated CubeSat system concepts

Fractionated spacecraft allocates mission functionality to multiple flight units, which interact wirelessly with each other to deliver on mission goals [17]. Benefits brought in from fractionation come primarily from an uncertainty management perspective, with a real options approach. A fractionated architecture allows for flexible adjustment of satellite capacity through incremental addition of spacecraft modules over the lifetime of the mission. Fractionated spacecraft were originally proposed by DARPA with the F6 program, which was later canceled. The seminal paper by Brown and Eremenko proposed a technology demonstration of fractionated spacecraft on microsatellites. They envisioned technology challenges in cluster flying, data transmission, fractionated navigation, distributed computation and data resources, power transmission, and force and torque transmission [17]. Fractionated architectures for future CubeSat implementations may include hybrid systems where heterogeneous spacecraft of different sizes interact wirelessly in a functionally dependent configuration. A fractionated architecture could include a large "mothership" spacecraft with a set of "daughter" fractionated CubeSat appendices for distributed observations or advanced geostationary (GEO) communication systems. The latter are akin to GEO clusters that have been recently proposed in the scientific literature [18]. While some of these concepts may seem far away in the future, they offer the key advantage of flexibility for modular capacity expansion. Functional interdependence also allows developing fractionated systems with low cost units, each of which able to optimize platform performance to the specific needs of the payload. A fractionated approach also allows for diversifying service offering by allowing for flexible interchangeable

payload fractions. Despite the ambition of the technical idea and its high impact in the scientific literature, no fractionated spacecraft is being flown or developed to date, at least in the public domain.

2.1.4 Federated CubeSat system concepts

Federated spacecraft systems are a more recent concept of distributed satellite systems following the idea of shared economies [19] as seen on terrestrial applications for public transportation. Federated satellites are heterogeneous, independently operated systems that interact in orbit on a need basis. Interaction is enabled by intersatellite links, distributed computing approaches, and delay-tolerant network protocols [20]. Optical communications have been identified as a key enabling technology for federated systems. Free-space laser communications allow mitigating interference and spectrum saturation traditionally associated with radiofrequency (RF) cross-links among large numbers of orbiting spacecraft. Given the complex orbital geometries and cross-link configurations that are envisioned for federated systems, optical communications appear a key enabling technology for the concept.

The key advantage of satellite federations is the opportunity to provide operators the ability to “resell” part of their unused capacity or reconfigure their spacecraft to serve other mission needs in case of changing operational requirements or partial system failures. The advantages and limitations of federated spacecraft systems are very much akin to those of cloud computing infrastructure on Earth [21]. In-space cloud computing infrastructure has been proposed in the context of federated systems [22]. Potential applications of CubeSat federations include the ability of deploying “virtual satellite missions” [23] by coordinating the use of orbiting resources that are made available by satellite operators to a centralized coordinating infrastructure. The coordination infrastructure can also be decentralized through modern in-space blockchain approaches [24]. Auctioning systems have been proposed to regulate the operations of federated systems in orbit. Applications in the domain of Earth observation have been recently explored in a value chain analysis of the European Copernicus Earth observing infrastructure [25].

Federated systems represent a valuable opportunity for CubeSat systems. By operating in federations, CubeSats can tap in otherwise unaffordable computing and processing resources. These are typically out of reach of CubeSat systems due to the stringent SWaP requirements associated with CubeSat platforms. Previous research has explored the technical and commercial feasibility of customer spacecraft operating federated operations with the International Space Station (ISS) as a supplier node [19]. One could therefore conceive CubeSats as low cost, agile data gathering nodes, using the ISS or other large spacecraft for centralizing complex computations, data interpretation, and so on.

Federated CubeSat systems have not yet been demonstrated in orbit. A first technology demonstration is slated to fly in 2019 [6]. Federated CubeSat concepts and related enabled services are picking up traction in the space industry. A recent paper by ESA analyzed the concept of “\$99 satellite” or “SpaceBNB” of nanosatellites, which closely align to the concept of spacecraft federations [26].

3 Enabling technologies

Distributed CubeSat concepts are bringing new technology challenges related to coordination and interaction among flight systems. [Table 3](#) provides an outline of selected enabling technologies for the distributed CubeSat system concept survey in [Section 2.1](#). The technologies outlined in the table are being developed (or will be developed) for many platform sizes, including nanosatellites. CubeSats bring to those challenges additional constraints in terms of size, weight, and power (SWaP) typical of the CubeSat standard.

Table 3 Overview of selected enabling technologies (in alphabetic order) for distributed CubeSat systems.

Technology	Enabled capability
Blockchain	Secure ledger to regulate data exchanges within a distributed CubeSat system. For example, to enable resource exchange marketplaces (for downlink, data storage, data processing, and other resources) in satellite federations
Capacity request forecasting and scheduling using deep neural networks	Regulation and interoperability coordination mechanism within DCSs with uncertain topology and uncertain demand (e.g., satellite federations)
Dynamic platform-distributed resource allocation and balancing	Operational coordination of satellite platform resources within DCSs
Explainable artificial intelligence (AI) for platform management and intelligent payload operations	Enabler for the use of AI in safety-critical (or reliability-critical) applications
High-accuracy CubeSat attitude control (<0.01 degrees) with high agility	Close-distance formation flight and high performance payload operations enabler (e.g., highly directional high-frequency antennas)
Higher efficiency/longer lifetime in miniaturized electric propulsion systems for CubeSats (e.g., next-generation ion electrospray thrusters [27])	Orbital station-keeping and plane change corrections
In-orbit spacecraft identification	Space traffic management and reliable in-orbit commissioning in rideshare launch missions
In-space Internet connectivity	Ground stationless missions, relying on common multimission ground system infrastructure operating the Internet in space
Magnetic torque and force transmission [17]	Orbital reconfiguration of fractionated satellite systems

Table 3 Continued

Technology	Enabled capability
Optical communications for CubeSats (intersatellite link and downlink)	Higher-performance communication capabilities with lower specific power consumption (in terms of W/kg) and with lower interference issues License-free communications
Out-of-plane reconfigurable intersatellite links	Enabling DCSs with complex orbital topologies (e.g., opportunistic networks between heterogeneous spacecraft nodes)
Power-efficient software-defined radio	Flexible and interoperable communication capabilities
Satellite quantum key distribution [28]	Secure in-space networking
Relative navigation	Formation flight
Satellite negotiator [29]	Interoperability with legacy spacecraft
Software-defined satellite	In-orbit reconfigurability of mission objectives
Space traffic monitoring and collision self-avoidance [30]	Secure deployment of large number of spacecraft
Virtual space missions [23]	Customer service offering of very lost cost mission capabilities

4 Conclusion

CubeSats have evolved from being systems designed for pure educational use to advanced satellite platforms for operational use in Earth observation, telecommunications, science, and other applications. CubeSats are of particular interest to the evolution of commercial space ventures due to the lower capital costs associated with their development and operations compared with traditional missions. The evolution of CubeSats has been largely facilitated by the exponential evolution of miniaturized technology and consequently the enabled advanced capabilities for those platforms. Distributed concepts and architectures are a similar evolution of CubeSat capabilities akin to the evolution observed in larger satellite platform systems in the past. Distribution in the form of constellations, formation flight, swarms, fractionation, and federations entails the capabilities of interaction and coordination among flight units. By means of distributed capabilities, CubeSat missions can effectively complement large institutional infrastructure or offer new instrument capabilities in space applications. Many (if not most) of the concepts described in this chapter—except for constellations—have not yet been demonstrated in flight. Distributed systems offer new opportunities for research and development and demonstration of new mission capabilities in orbit. The “distributed revolution” has started with the launch of operational CubeSat constellations. Given the opportunities they can provide at cost, it is likely to expect that other distributed CubeSat systems will appear in orbit in the future.

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